Determination of the Young modulus of PDMS polymer from Atomic Force Microscopy

Alessandro Ciavatta, Elisa Damiani, Riccardo Gaeta

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Abstract

The Hertzian model of contact mechanic is used to first calibrate the AFM and then to estimate the Young modulus of a slice of PDMS via a AFM force vs. indentation analysis. The AFM sensitivity is calibrated by performing a force spectroscopy analysis on a hard glass surface while the cantilever spring constant has been measured by the thermal tune method. The spectroscopy profiles used to extrapolate PDMS's Young modulus are hysteretic: there is a dependence on the scanning direction. The best estimations are an elastic modulus of 4 MPa during approach and 5 MPa during retraction. An amplitude/distance spectroscopy is performed but from the data it was not possible to estimate with certainty the origin and the intensity of the force gradients.

Introduction

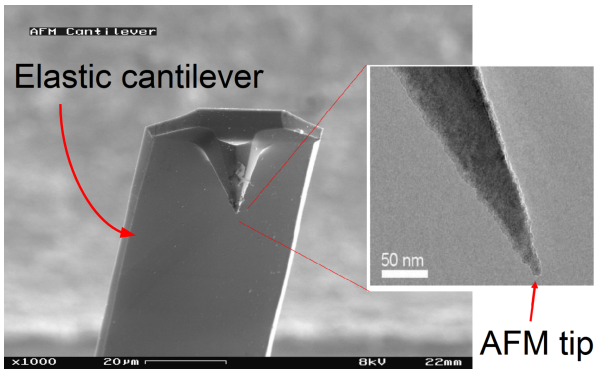


Figure 1: Electron microscope image of an AFM cantilever and tip

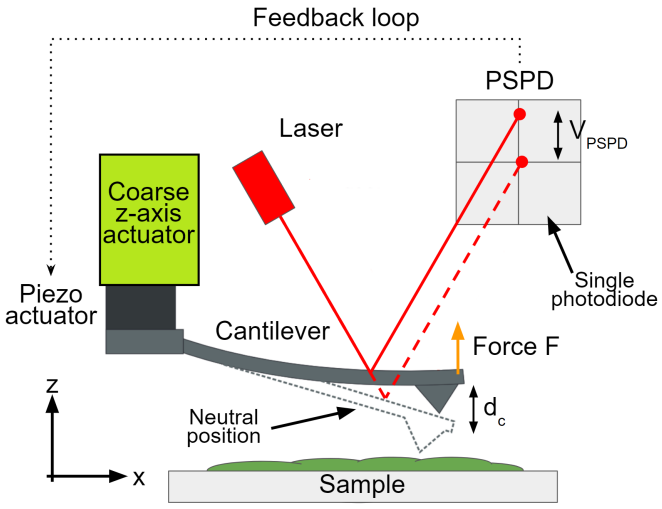


Figure 2: Schematic functioning of an AFM

An Atomic Force Microscope is a type of microscope that uses touch instead of light. A microscopic tip made out of Silicon (or SiN) is used to sense the surface of a sample exploiting the various forces felt at the nanoscale. In Fig.(1) an image of a nanometric-sharp conical AFM tip can be seen.

To infer the position of the cantilever a laser and a position sensitive photo detector (PSPD) are used. In Fig.(2) an exaggerated example of how four photodiodes can measure the position of the cantilever is shown.

As the cantilever bends upwards the diodes in the top part of the PSPD are more and more illuminated creating a differential voltage signal V_{PSPD} that the instrument uses to determine the position of the tip. The neutral position of the cantilever is the position that the cantilever assumes when no force is acting on the tip. When

a tip indents (or gets close to) a surface, the cantilever will bend and the reflected laser light will shift on the PSPD. The PSPD sensor is composed of four photodiodes to also measure any lateral deflection that the cantilever might experience.

To move the tip towards the surface two actuators are used: one coarse that has a resolution of $\sim$1 µm and a piezoelectric actuator (piezoscanner) that can reach 0.01 nm resolution.

To move the sample in the plane an XY-scanner is present under the sample.

For small displacement the V_{PSPD} signal can be assumed linear with respect to the cantilever deflection d_c as the deflection itself can be considered linear to the force F acting on the tip. The relations between V_{PSPD}, d_c and F are:

$$V_{PSPD} = S \cdot d_c \quad \text{and} \quad F = k \cdot d_c \tag{1}$$

where S is the sensitivity of the instrument and k is the spring constant of the cantilever.

An AFM is not only useful for nanoscale topographic imaging, by dragging the tip close to (or in contact with) a sample surface, but it's also a valid tool for the determination of elastic properties of materials. To measure the Young modulus of a material the tip is brought in contact to the sample and is pushed against it. The tip will indent the material and it will experience a repulsive force that depends on the indent distance δ.

The Hertz formula that correlates the repulsive indentation force to δ, for a conically shaped tip is:

$$F = \frac{2}{\pi} \frac{E \tan\theta}{1-\nu^2} \delta^2 \tag{2}$$

where E is the elastic Young modulus of the sample, θ is the cone angle of the tip (nominal value 9°) and ν is the Poisson ratio (0.5 for a perfect elastomer). This analysis, conducted in contact mode (CM) in the repulsive force region, is called force-distance (F/d) spectroscopy. While close to the surface, the tip usually feels a strong attraction and snaps in contact. This is detected as a discontinuity in the force vs. δ graph. During retraction, instead, the adhesive forces will keep the tip bound to the surface (pull-off) for longer.

In non-contact mode (NCM), the base of the cantilever is instead constantly oscillating at a known frequency, while the response of the tip is monitored through the PSPD signal. In this mode, amplitude distance (a/d) spectroscopy can be performed by keeping track of the response of the amplitude as the tip approaches the surface. For small amplitude oscillations the system can be modeled as a simple forced harmonic oscillator. There are two external forces acting on the tip: the piezo-vibrator driving force and tip-sample interactions. The tip-sample interactions can be expanded up to first order and be considered linear within the tip oscillation amplitude.

Keeping piezo-vibrator angular frequency ω fixed, oscillation amplitude response A in relation to the tip-sample distance z is:

$$\begin{aligned} A(z,\omega) &= A_0[(1-(\omega/\omega_0)^2 - F'(z)/(m\omega_0^2))^2 \\ &\quad + (\omega/\omega_0)^2/Q^2]^{-1/2} \qquad (3) \\ &\simeq A_0|1-(\omega/\omega_0)^2 - F'(z)/k|^{-1} \qquad (4) \end{aligned}$$

where ω_0 is the oscillator's resonant angular frequency, Q the quality factor, $k = m\omega_0^2$ the spring constant and F' the tip-sample force gradient; the approximation is valid for $Q \gg 1$.

The aim of these work is to use the Hertzian model of contact mechanics to calibrate the AFM and measure the Young modulus of a sample of polydimethylsiloxane (PDMS): a silicon-based elastomer.

Materials and methods

The AFM used in this experiment is the Park NX10 [1]. At the time of the experiment the XY-scanner was broken and so the AFM is used to perform force measurements and force spectroscopy. The resolution of the AFM depends strongly on the magnitude of the external noise so, to reduce mechanical vibrations, such as sound waves, the AFM is set on a vibration cancelling table.

A microscope, mounted onto the AFM, is needed to calibrate the instrument: the position

sensitive photo detector (PSPD) laser needs to be centered on the cantilever and the mirrors of the AFM moved, so that the maximum signal comes at the center of the PSPD.

During this experiment, we are only interested in motion along the z-axis (height), so we are interested only in the difference between the signal recorded by the two vertical photodetectors (V_{PSPD} or A-B signal).

An important quantity is the Z height, that is the position of the base of the cantilever with respect to the coarse motor reported by the instrument. The Z-scanner, which controls the height, can be retracted and approached to the surface with a maximum elongation of some µm. The surface is approached first manually down to around 2 mm then automatically.

The spring constant is typically calculated from the response spectrum of the cantilever tip to white thermal noise. Room temperature is enough to perform such measurement, it's sufficient to record the PSPD signal of the free cantilever over some seconds. To calibrate the Z-scanner position with the PSPD signal, and obtain the sensitivity S an F/d analysis is carried out on a hard glass substrate that is considered ideally rigid.

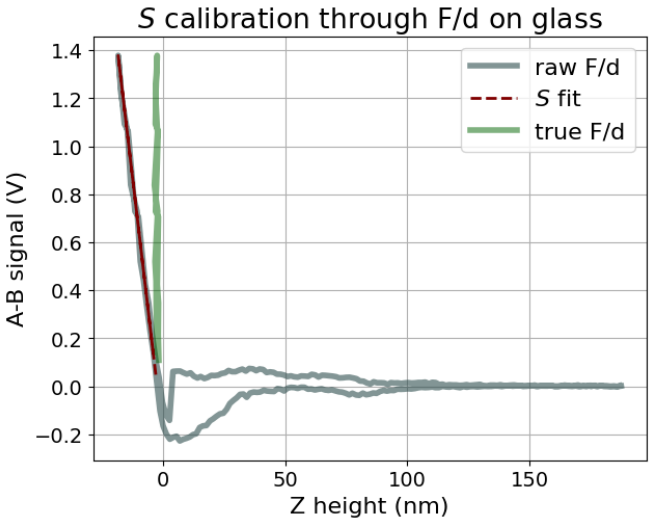


Figure 3: Sensitivity fit for F/d spectroscopy data on glass, used to track the cantilever's tip position from the A-B signal

Results and discussion

The sensitivity S is estimated from F/d spectroscopy data on a glass slide: an ideally impenetrable material. In Fig.(3) the sensitivity is given by the slope of the linear fit in the contact region. A small displacement of the base of the cantilever, displayed as Z height coordinates, is accompanied by an equal deflection of the tip, sensed by the A-B signal. Sensitivity is thus estimated as $S = 86.4$ V/µm.

The frequency response of the free cantilever's tip to thermal noise (Fig.(4)), is used instead to calculate the spring constant k. From the equipartition theorem,

$$\frac{1}{2}k\langle d_c^2\rangle = \frac{1}{2}k_B T,$$

where $\langle d_c^2\rangle$ can be simply calculated from the variance of the PSPD signal and the sensitivity, since $d_c = V_{PSPD}/S$. At these frequencies, we expect a completely white perturbation spectrum, hence the response is expected to be the (squared) resonance curve of the harmonic oscillator. We notice from the fit that a white noise offset is present in the PSPD signal, which is probably associated with a calibration offset in the A-B signal and should not be taken into account in the estimation of the integral variance,

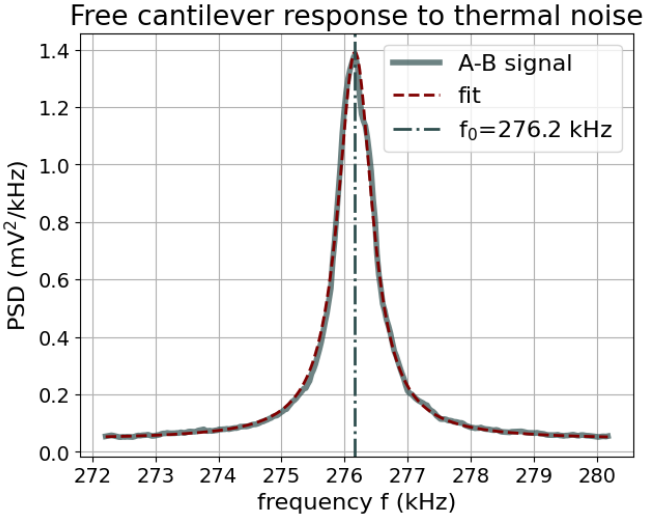


Figure 4: Power spectrum of the A-B signal, during thermal tuning. The signal is fitted with $A(\omega)^2$ from Eq.(3) for free oscillations ($F' = 0$)

given directly by the other fit parameters.

We notice also that this method seems more effective than the sweep method (Fig.(5)) in determining the dynamical properties of the oscillator, such as the angular resonant frequency ω_0 and the quality factor Q: the resonant peak is more symmetric and less noisy.

Results of thermal tune method are: $Q = 415$, $f_0 = 276.17\,\text{kHz}$ and $k = 11.0$ N/m, assuming $k_B T = 26$ meV.

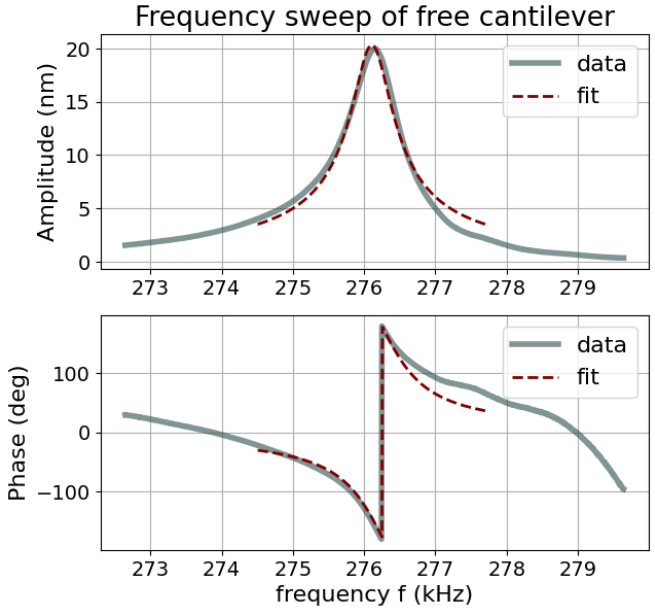


Figure 5: Response curve of the cantilever's tip to a frequency sweep of the piezo-vibrator

After this analysis, we have reconstructed a true F/d profile on PDMS. Results are shown in Fig.(6). A-B signal is transformed to deflection d_c through the sensitivity and then to force using the elastic constant k. Eventual offsets in A-B signal, due to poor calibration, can be easily removed by shifting the free cantilever part of the F/d spectrum ($\delta > 200nm$) at $F = 0$. Z height is then recasted as indentation $\delta = Z + d_c - Z_0$, where Z_0 is an arbitrary offset, usually defined as the point after the snap-in where the forces come back to zero.

Due to the presence of hysteresis, two different fit are performed, in the approach and retract region. The results, unfortunately, depend strongly on the fit range and on the region: from the fit range indicated in Fig.(6), the Young

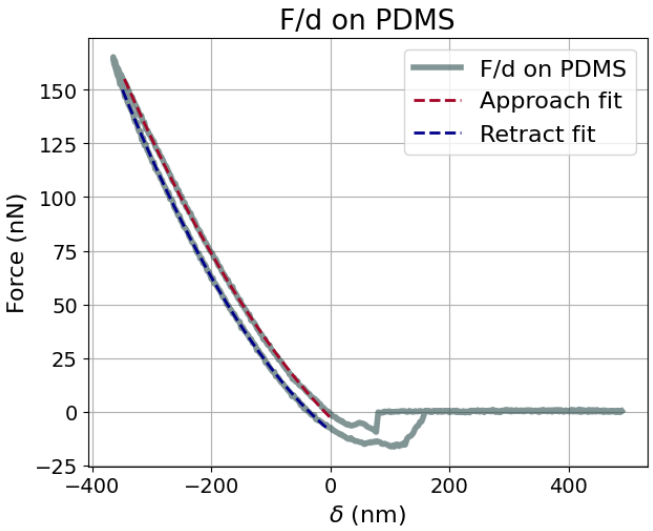


Figure 6: F/d profile of PDMS polymer. Near $\delta = 0$ we can observe the "snap-in" and "pull-off" behaviours. The fit is parabolic (Hertz model); two fit are performed because of the hysteresis of the elastomer

modulus of the PDMS sample is estimated at $E = 4\,\text{MPa}$ during approach and $E = 5\,\text{MPa}$ during retract. With other fit ranges we obtain different E, but always of the order of the MPa. Literature values of the Young modulus of PDMS range from 0.36 to 0.87 MPa [2], about one order of magnitude lower than our results. This difference could be due to inhomogeneities of the PDMS. The first correction we suggest is a geometrical one: we cannot directly know the shape and the integrity of the tip without doing SEM on the tip itself. With a different geometry, for example with a four-sided pyramid tip of the same angle θ we would obtain a 10% lowering of the elastic modulus [3]. Moreover, the two curves experience a polynomial growth weaker than quadratic. Even this anomaly can be explained also with different geometries. Fluctuations of E varying fit ranges can be due to interactions between the tip and the substrate beneath the PDMS thin film, or again inhomogeneities of the PDMS. The indentation velocity also plays a role in the hystereretic behaviour [3]. If the spectroscopy is conducted too fast, viscous forces become predominant over the elastic ones, invalidating the Hertz model; however, with too

small velocities the material would be deformed plastically, resulting in a mismatch between the approach and retract curves.

A/d spectroscopy

Disclaimer: our a/d data turned out to be not satisfactory since the tip was apparently too close to the surface and got stuck to it. We wish to thank Group 3 for providing this data.

During NCM, the operation frequency is set to a frequency off-peak, but near, where the slope of the resonance curve is maximum, i.e. the sensitivity is maximum. For this reason, before measuring in NCM a frequency sweep on the free cantilever is performed.

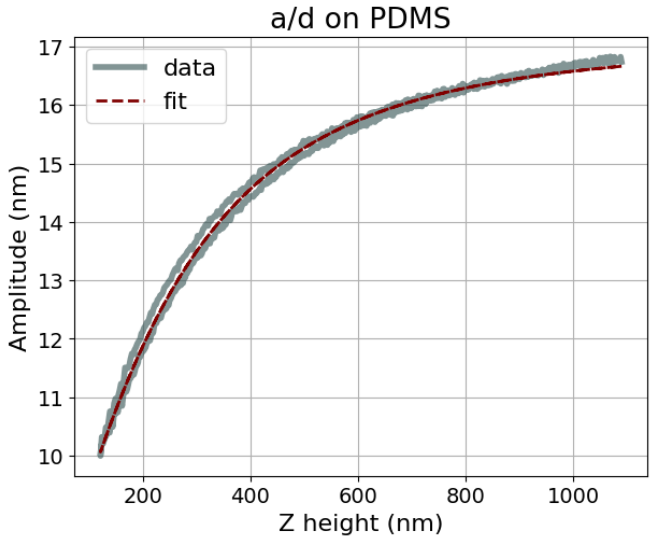


Figure 7: Amplitude-distance profile, fitted assuming a force gradient $F' \sim (z - z_0)^{-3}$, coherent with the behaviour of Van der Waals attractive forces

The results of the a/d spectroscopy shown in Fig.(7). Operating at a fixed angular frequency ω, we can relate the damping of free oscillations at driving amplitude once the surface is reached. With such small oscillation amplitudes (under 20 nm), such large Q-factor ($Q = 415$) and with operating frequencies ω very near ω_0 (ω/ω_0 of the order of unity), the approximation in Eq.(4) is valid. The sign of the force gradient cannot be extrapolated, since ω/ω_0 is unknown; therefore we cannot distinguish between repulsive and attractive forces. Assuming, from the previous Hertz model, a quadratic repulsive force, we would expect a linear force gradient. This is in contrast with our data, which does not resemble a $|z-z_0|^{-1}$ curve. Fit in Fig.(7) is performed under the assumption of an attractive force $F \sim -z^{-2}$, like the total Van der Waals force. From fit results, we would find a rough estimation of Hamaker contant A_H for SiO_2-PDMS: the product $A_H \cdot (1-(\omega/\omega_0)^2)$ is of the order of 10^{-10} J, to be compared with $A_H = 5.4 \times 10^{-20}$ J [4]. In order to confirm Van der Waals forces hypothesis, ω should be unreasonably close to ω_0. A/d curve fits well with $F \sim -z^{-\alpha}$ for a large interval of α, therefore forces could in principle have an electrostatic origin: PDMS is non-conductive, hence fixed charges at the surface can polarize the tip, giving rise to charge-dipole interaction.

Conclusions

We have explored the two operation modes of AFM: contact mode (CM), through force/distance spectroscopy (F/d) and non-contact mode (NCM), through amplitude/distance spectroscopy (a/d). We have calibrated the instrument by performing F/d on a hard substrate and by thermal tuning. Contact forces have been directly measured for a PDMS thin film elastometer, through F/d spectroscopy. Comparing with Hertz model, we have estimated Young modulus of the elastomer: $E = 4 - 5$ MPa, to be compared with $0.36-0.87$ MPa found in literature [2]. From a/d spectroscopy data we could not estimate with certainty the origin and the intensity of the force gradients.

References

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